



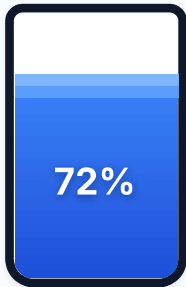
ST. XAVIER'S COLLEGE (AUTONOMOUS)
Ahmedabad, Gujarat, India

• SMART WATER MANAGEMENT

Smart Water Tank Monitor

An automated **IoT system** designed to prevent water overflow, save electricity, and provide real-time updates directly to your devices.

HC-SR04



LIVE LEVEL

Core Functionality

This unit uses non-contact ultrasonic waves to detect the exact liquid levels in your tank. An intelligent automation controller monitors custom low and high thresholds, automatically switching a relay to manage the water pump. You get instant local alerts via a built-in buzzer and online visibility on a live web dashboard.

NodeMCU WiFi

Ultrasonic Sensor

Relay Auto-Control

Piezo Buzzer Alert

Web Dashboard API

24/7

CONTINUOUS MONITORING

±1 cm

SENSOR PRECISION

100%

AUTOMATIC OPERATION

0

MANUAL EFFORT

UTILITY IN DAY-TO-DAY LIFE



Home Automation

Ensures your household water tank is never empty by automatically refilling when levels drop below 20%.



Resource Conservation

Prevents tank overflows, stopping water wastage and significantly reducing monthly utility bills.



Instant Visibility

Provides immediate status readouts from a phone or computer, eliminating the need to physically climb up.



Agricultural Irrigation

Maintains optimal storage tank levels automatically for livestock watering and drip-irrigation systems.



Facilities & Commercial

Tracks water levels across clinics, offices, and multi-tenant blocks to ensure constant water supply.



Pump Protection

Shuts down the motor automatically when the source tank is empty, preventing pump dry-run damage.

TECHNICAL CAPABILITIES



Precision Sensor Reading

Uses non-contact HC-SR04 sonic distance ranging for exact height calculations, unaffected by water impurities.



Custom Control Thresholds

Define exact start and stop percentages inside the code to manage pump activation and shut-off cycles.



Local Alarm System

A secondary physical buzzer sounds immediately if critical overflow or underflow boundaries are breached.



Web UI & REST Interface

Transmits data packets over local Wi-Fi networks to sync a mobile-friendly dashboard in real-time.

OPERATIONAL WORKFLOW

1

Measurement

Sensor fires sound pulse and measures echo bounce-back time.



2

Computation

NodeMCU translates the delay to distance and level percentage.



3

Decision

Relay turns pump ON if <20% and OFF when level reaches >90%.



4

Telemetry

Sensor metrics package is posted over HTTP to the backend server.



5

User Portal

Status dashboard renders live statistics on your phone or PC screen.



Smart Water Tank Monitor

IoT Based Monitoring & Control System | College Project

Hardware Prototype

ESP8266 Project